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The effects of a mobile endless runner exergame on users daily step count.

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Zusammenfassung

Diese Studie untersucht die Auswirkungen eines selbst entwickelten mobilen 3D-Endless-Runner-Exergames auf die tägliche Schrittanzahl der Nutzer. Die Forschung beschäftigt sich vermehrt mit Exergames, welche körperliche Aktivitäten mit interaktiven Videospielen verbinden. Das Hauptziel dieser Studie ist es, festzustellen, ob die Verwendung eines mobilen Exergames zu einer signifikant erhöhten täglichen Schrittanzahl der Teilnehmer führt. Durch eine Kombination der quantitativen Methode, welche die täglichen Schritte ermittelt und der qualitativen Methode, welche in Form von Tagebucheinträgen Feedback liefert, wird eine umfassende Analyse des Nutzerverhaltens und der Nutzungserfahrung durchgeführt.

An der Studie nahmen zwei Teilnehmer (eine Frau, ein Mann) im Alter von 24 und 27 Jahren teil. Die Studie war in zwei Testphasen unterteilt, die jeweils vier Wochen dauerten. In der ersten Phase, der Vergleichsphase, wurde die tägliche Schrittzahl der Teilnehmer ohne Verwendung des Exergames überwacht. Im Anschluss folgte die zweite 4-wöchige Testphase, in der die Teilnehmer das Exergame verwendeten und das Tagebuch führten, wodurch ihre Erfahrungen dokumentiert werden konnten. Die Schrittanzahlen wurden kontinuierlich mittels Smartphone-Sensoren und -Anwendungen aufgezeichnet, um eine präzise Messung der körperlichen Aktivität zu gewährleisten. Ziel war es, das Gehen der Teilnehmer durch das Exergame zu stimulieren. Das 3D-Endless-Runner-Exergame wurde vor Beginn der Testphasen programmiert und getestet.

Die Datenanalyse umfasste deskriptive Statistiken, vergleichende Analysen, Trendanalysen sowie die Integration von qualitativem Feedback. Die Ergebnisse zeigten, dass die Steigerung der Schrittanzahl der Teilnehmer zwar nicht signifikant war, jedoch im Durchschnitt höher lag als ohne Verwendung des Exergames. Die Teilnehmer berichteten sowohl über positive als auch negative Nutzungserfahrungen sowie eine anfänglich erhöhte Motivation zur körperlichen Aktivität.

Die Ergebnisse deuten darauf hin, dass mobile Exergames als Mittel zur Förderung der körperlichen Aktivität durch eine spielerische Herangehensweise dienen können, insbesondere wenn sie die Langzeitmotivation der Nutzer erhöhen. Diese Studie unterstützt die Weiterentwicklung und den Einsatz von Exergames im Gesundheitsbereich und schließt mit einer Diskussion über praktische Implikationen, Einschränkungen und Vorschläge für zukünftige Forschung im Bereich Exergaming.

Abstract

This study investigates the impact of a self-developed mobile 3D endless-runner exergame on users' daily step counts. The research is increasingly focusing on exergames, which combine physical activity with interactive video games. The primary objective of this study is to determine whether the use of a mobile exergame leads to a significant increase in participants' daily step counts. A comprehensive analysis of user behavior and experience is conducted through a combination of quantitative methods, which measure daily steps, and qualitative methods, which provide feedback in the form of diary entries.

The study involved two participants (one female, one male) aged 24 and 27 years. The study was divided into two test phases, each lasting four weeks. In the first phase, the control phase, participants' daily step counts were monitored without the use of the exergame. The second four-week phase involved the use of the exergame by the participants, during which they kept a diary to document their experiences. Step counts were continuously recorded using smartphone sensors and applications to ensure accurate measurement of physical activity. The aim was to stimulate participants' walking through the exergame. The 3D endless-runner exergame was programmed and tested prior to the commencement of the test phases.

The data analysis included descriptive statistics, comparative analyses, trend analyses, and the integration of qualitative feedback. The results indicated that although the increase in step counts was not statistically significant, the average step count was higher with the use of the exergame compared to without it. Participants reported both positive and negative experiences with the exergame and an initially increased motivation for physical activity.

The results suggest that mobile exergames can serve as a tool for promoting physical activity through a playful approach, particularly if they enhance users' long-term motivation. This study supports the further development and application of exergames in the health sector and concludes with a discussion of practical implications, limitations, and suggestions for future research in the field of exergaming.

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1. Introduction

1.1 Purpose of the Study

The primary purpose of this study is to investigate the impact of a self-developed mobile 3D endless-runner exergame on users' daily step counts. This study highlights the health benefits of walking, a fundamental physical activity. Walking is an essential component for maintaining health and well-being. Regular walking improves cardiovascular health, enhances muscle endurance, and reduces the risk of chronic diseases such as diabetes, obesity, and hypertension[LL12]. Given the widespread sedentary lifestyle in today's society, it is increasingly important to motivate people to incorporate more steps into their daily routines. The health risks associated with physical inactivity can be mitigated by increasing daily step counts[LSL+12].

To address this issue, the study integrates physical activity through interactive exergames. The exergame is designed to provide an enjoyable experience for the user while playing. This fun and engaging format is intended to motivate participants to maintain their newly acquired walking habits. The exergame offers immediate rewards to the user in the form of fun, ambition, and other motivational triggers, which can help reinforce positive behavior changes. Such interactive games have the potential to change and sustain habits through entertaining and repetitive play, thereby helping users build routines that incorporate regular physical activity[CS12].

1.2 Background

The study is based on the increasing interest in the intersection of gaming and physical activity, particularly with the rise of exergames. Exergames are video games designed to promote physical activity through interactive gameplay. The concept of combining physical activity with video games gained prominence with the introduction of the Nintendo Wii, a motion-sensitive gaming system. These systems demonstrated that games could successfully motivate people to be more physically active[SHM07].

Physical inactivity is associated with numerous health problems, leading to reduced life expectancy. These health issues include heart disease, type 2 diabetes, and certain

types of cancer. Studies have shown that increasing physical activity can prevent a significant portion of these diseases and extend life expectancy[LSL+12]. This study focuses on a mobile 3D endless-runner game, a specific type of exergame. The game is designed to be played on a smartphone, making it more accessible, practical, and easier to integrate into daily life. The primary objective is to determine whether the use of this exergame can lead to a significant increase in users' daily step counts.

The study uses a mixed-methods approach, collecting both quantitative data (daily step counts) and qualitative data (user experiences documented through diaries). Through comprehensive analysis, the study aims to provide a detailed understanding of how the exergame influences physical activity and user motivation. By examining the effects of the mobile 3D endless-runner game, this study seeks to contribute to the promotion of users' health and fitness. The results are expected to provide insights into how exergames can be used to encourage more active lifestyles and address the challenges of physical inactivity.

2. Literature Review

2.1 Exergames and Physical Activity

2.1.1 Considerations for the design of exergames

The scientific paper “Considerations for the Design of Exergames” was written by Jeff Sinclair, Philip Hingston and Martin Masek from the School of Computer and Information Science at Edith Cowan University. The work deals with the study and design of exergames, which are a combination of video games and exercise equipment. The connection between video games and physical activity is a relatively old idea that has received increased attention since mid-2006.

One reason for the increased interest is the concern about the high obesity rates in Western societies, especially among children. Video games hold a fascination for children, which can be used to encourage more motivation for physical activity. Another reason why exergames are becoming increasingly popular is the financial potential. In the USA, training equipment worth 5.2 billion dollars was sold in 2007. Exergames are seen as potential growth area that could increase these figures further. With the release of the Nintendo Wii in 2006, there was an increased interest in exergames. The motion-sensitive controllers were advertised as a means of promoting physical activity.

Despite its success to date, the history of exergaming has been mixed. In order to understand why some systems have been successful and others not, the paper examines the history of exergaming and analyzes the current state of research in this area from 2007. A central model for explaining the success of video games is Csikszentmihalyi’s "flow" concept, which can be applied to exergaming. This model describes how the balance between a player’s skills and the challenges in the game leads to an optimal state of engagement.

The success of video games can be described using the “flow” concept. It is a central model that was developed by Csikszentmihalyi. This model can also be applied to exergames and it explains how the optimal balance between a player’s skills and the challenges in the game leads to a state of maximum motivation and concentration.

The authors recommend extending the flow concept from the psychological level to the physical fitness level, incorporating proven methods from fitness program design to enable more effective and engaging training.

In advanced countries, the obesity rate among children and adolescents is increasing. It has increased significantly in the last 30 years in particular. A negative example is Australia, where the overweight and obesity rate among children and adolescents is 33%. These worrying figures have increased interest in exergaming as a possible solution. For exergames to be successful, it is essential that they are both engaging and effective. This means that the game should not only be entertaining and exciting, but also serve as an efficient training tool. This is the only way it can be enjoyable for players and support their fitness goals at the same time.

An innovative approach to improving exergaming systems is the "dual-flow" model, which considers both psychological and physiological factors. The appeal of the game is illustrated by the classic flow model, which describes the immersion and excitement of the players. At the same time, the effectiveness of training is ensured through the careful balance between physical fitness and training intensity to achieve optimal health benefits. Understanding and applying these models could be critical to making future exergames more successful and maximizing both game quality and health benefits. Exergaming has the potential to play an important role in tackling obesity while maintaining enjoyment of the game. [SHM07]

2.1.2 Turning the Classic Snake Mobile Game into a Location–Based Exergame that Encourages Walking

The paper “Turning the Classic Snake Mobile Game into a Location-Based Exergame that Encourages Walking” by Luca Chittaro and Riccardo Sioni examines the transformation of the classic Snake game into a location-based exergame called LocoSnake that controls players by walking. This exergame is designed to combine physical exercise and gaming to promote physical activity and health. The use of mobile devices for exergames is becoming increasingly popular. Outdoor activities such as walking and running can be monitored and supported more easily. The goals of LocoSnake are to motivate the user to do more moderate physical activity outdoors. In addition, the attitude towards walking should be positively influenced.

In the study conducted, generally recognized evaluation tools from the medical literature were used to thoroughly and precisely record and evaluate the effects of the exergame on the users. Users’ lifestyles have an impact on their attitude towards exercise. This can also influence the enjoyment of the game. One important point that was determined with the help of the study is that the walking speed of LocoSnake players was higher. This was compared with the group that walked the same route without playing the game.

The increased walking speed of the users was particularly noticeable when the fruits in the game were further away. This result indicates that it is a strategic adjustment of speed. There was further feedback indicating that precise and targeted input is very important. The users had the feeling that the snake was too imprecise to control. This result was also reflected by low scores in feedback and control.

For people with a sedentary lifestyle, the game was surprisingly successful. People who lead a physically active lifestyle found walking less pleasant due to the increased

concentration while playing. The authors underline the urgency of further research to confirm the applicability of the results. In particular, they emphasize the need to include female participants and to consider a variety of other exergames in order to obtain a more comprehensive picture.

These additional studies could provide valuable insights and strengthen the generalizability of the current findings. Social elements that allow information to be shared with friends could add more depth to the game and also increase the interest of more physically active people in the game.

In summary, it can be said that LocoSnake has the potential to motivate people to be more physically active. The attitude towards walking is also strengthened, which was confirmed by the study. The conclusion from the data collected indicates that a well-designed exergame can lead to an improvement in people's health. The positive effects are significantly stronger in individuals with a lower activity level. [CS12]

2.2 Mobile Endless Runner Games

2.2.1 Activity Recognition for Movementbased Interaction in Mobile Game

This thesis is dedicated to the development of an innovative 3D endless runner game called ActivRunner. Integrating an advanced activity recognition system, the game recognizes and predicts four different movements in real time: standing, moving left, moving right, squatting, and jumping. The project aims to replace conventional touch operation with a more user-friendly and natural control system using movements. This method is intended not only to create new, immersive mobile gaming experiences, but also to promote physical activity among players.

ActivRunner thus represents a significant step towards innovative game development with a focus on user-friendliness and health promotion. The authors argue that smartphones offer great potential for motion-based interactions due to their large number of sensors, such as accelerometers and gyroscopes. Most mobile games only use augmented reality and tracking, which is location-based. ActiveRunner uses motion detection on a deeper level. The study demonstrates that such systems are not only technically feasible, but are also able to involve users more in the game and motivate them to be more physically active.

In the data collection phase, data was gathered from six volunteers in order to train the activity recognition system. The volunteers practiced the required movements with their smartphones in their hands. The data is collected in packets of 64 values. This collected data was recorded at a frequency of 50 Hz, which corresponds to movements of 1.28 seconds. This data was then used to train and test the system in real time.

An important component of the game is the prevention of cheating. Each participant was instructed to try to cheat during the training phase. They should make hand movements during the standing activity to simulate active movements. This is intended to arm the system against attempts to cheat. This increases the chances that players will perform the correct body movements instead of trying to trick the system, as the movements are only tracked during real activity.

ActiveRunner is an endless runner game. The character runs through the world on a three-lane track, using body movements to avoid obstacles. The game requires a smartphone and is controlled by body movements. The only interaction required is for the players to adjust their body positions and movements.

The results of the experiments showed that the system is both robust and responsive. The participants praised it in particular for its ability to successfully fend off manipulation attempts. In addition, the immersive experience and the high level of physical activity that the game enables were rated particularly positively.

In their future work, the authors intend to develop more elaborate and versatile games. They also plan to carry out more extensive tests and data collection. These measures should help to further optimize the accuracy and performance of the system. In addition, they want to investigate the extent to which a single smartphone is sufficient to accurately predict different activities and whether anti-cheat training is feasible with the introduction of more activities.

Overall, this study shows that combining mobile games with body movement is an effective and fun way to promote physical activity, especially among younger target groups. [AA17]

2.3 Mobile- and Wearable-Sensor accuracy

2.3.1 A Survey on Human Activity Recognition using Wearable Sensors

The study "A Survey on Human Activity Recognition using Wearable Sensors" by Óscar D. Lara and Miguel A. Labrador provides a comprehensive overview of the current state of the art in the field of Human Activity Recognition (HAR) using wearable sensors. The focus is on accurate and timely detection of human activity and behavior, which is of great importance in various fields such as medicine, security, entertainment and the military.

The paper describes a general architecture of a HAR system and explains the main components that must be present in any such system. The authors categorize HAR systems into learning approach and response time. The learning approach can be either supervised or semi-supervised and the response time offline or online.

They discuss the main challenges in this area, which are the accurate and timely detection of activities and the minimization of energy consumption. The solution presented in the paper emphasizes the use of an efficient machine learning algorithm and energy-saving sensors.

The study qualitatively evaluates 28 different systems in terms of recognition performance, energy consumption, obtrusiveness, and flexibility, among others. A key point of the study is the importance of feature extraction and machine learning techniques for the development of recognition models. Once the model is trained, new data instances can be evaluated to make predictions about the activities performed.

Outstanding approaches include the use of accelerometers and other sensors to capture motion data. The authors emphasize the need for further research in areas such as improving recognition accuracy, reducing energy consumption and minimizing

the intrusiveness of the sensors. Very large or obvious sensors could be perceived as disturbing, so an inconspicuous small sensor would be an improvement.

They also identify open problems that should be addressed in future research to revolutionize how people interact with mobile devices, such as the development of context-aware applications that are able to appropriately consider and respond to context, as well as environmental information and user activities. Similarly, the integration of environmental sensors that detect various environmental parameters such as temperature, humidity and lighting conditions is a challenge that still needs to be solved in order to enable a more comprehensive use of sensor data.

HAR systems offer significant opportunities in healthcare, especially for patients with long-term health conditions such as diabetes, obesity or heart disease who need to perform regular physical activity. Similarly, these systems can be used in the mental health field to monitor behavioral patterns and detect and prevent unwanted activities in patients with dementia and other mental health conditions at an early stage.

The military sector is also a conceivable area of application. Soldiers' information can be analyzed, such as activities, locations and health conditions, which can improve performance and safety. The sensors can also be used to support decision-making, which is particularly useful during training or combat

The study provides a detailed overview of progress to date and points the way forward for future developments in this dynamic field of research. Overall, the study shows that although human activity recognition has been an active field of research for over a decade, crucial aspects still need to be addressed in order to make significant progress and fundamentally change the way people interact with wearable technologies. [LL12]

2.3.2 Human Activity Recognition using Inertial, Physiological and Environmental Sensors: a Comprehensive Survey

The article "Human Activity Recognition using Inertial, Physiological and Environmental Sensors: A Comprehensive Survey" by Demrozi et al. discusses human activity recognition (HAR) using inertial (e.g., accelerometers and gyroscopes), physiological (e.g., heart rate monitors) and environmental sensors (e.g., security cameras, temperature) and provides a comprehensive overview of publications on HAR that focus on machine learning (ML). HAR has become increasingly important over the last ten years due to the proliferation of electronic devices such as smartphones and smartwatches.

This technology is used in various sectors, including healthcare, interactive games, sports and surveillance systems. The algorithms developed by HAR aim to recognize human movements in controlled and uncontrolled environments. They face several challenges that need to be solved. Everyday human movements are very complex and varied. The same activities can also differ.

In addition, a decision must be made on the balance between data protection and performance, and the devices used can also pose a challenge, as not every wearable

or embedded device has the same computing power or runs on the same operating system. The variety of data can also lead to difficulties in data annotation.

The data required for training and testing HAR algorithms usually comes from two main sources: On the one hand, from environmental sensors, such as security cameras, which provide comprehensive information about the environment. On the other hand, from embedded sensors, such as accelerometers, which are integrated into smartwatches and record detailed movement data of the wearer. These two types of sensors complement each other and provide a comprehensive database that makes it possible to precisely detect and analyze human activity.

The requirements for using deep learning (DL) in HAR are large amounts of data and high computing power to interpret this data. These sufficient data quantities and the necessary computing power cannot be found in the application areas of HAR. Deep learning (DL) is a subset of machine learning and is used to automatically learn and extract patterns and features from data.

Among older people, HAR has been classified as a promising tool to support their daily lives. The authors discuss the use of different sensor types and the application of deep learning methods to improve HAR accuracy. They emphasize that HAR systems need to be continuously developed to accurately capture and classify the diversity of human activities.

In summary, the article provides a comprehensive review of the current state of HAR research and shows how ML and DL can be used in combination with various sensors to accurately detect and monitor human activities. The challenges and limitations in this area are discussed as well as future research directions aimed at improving the efficiency and accuracy of HAR systems. [DPBR20]

2.3.3 A systematic review of smartphone-based human activity recognition methods for health research

The systematic review by Straczkiewicz et al. deals with the detection of human activity for health research using smartphone sensors. Due to their widespread use and integrated sensors, such as accelerometers and GPS, smartphones are particularly well suited for the continuous monitoring of everyday activities.

The authors identified 108 relevant articles that could be used for a systematic review. These articles were extracted from the Scopus, PubMed and Web of Science databases. These studies investigated the different approaches to activity recognition. The approaches can be categorized into four phases: data acquisition, data preprocessing, feature extraction and activity classification.

To collect data, they used commercially available smartphones. These devices, equipped with various sensors, were usually kept in trouser pockets, jacket pockets, bags or simply held in the hand. The studies reveal that smartphones are well suited to recognizing activities such as walking, running, climbing stairs, cycling and various household activities.

Various algorithms and machine learning methods are used to classify the activities. A key point of the review is the need to improve the quality of the collected data and to address missing data.

For future studies, the authors recommend a greater variety of different participants and activities to be carried out. Larger and more diverse study populations allow for more robust and generalized data needed to draw conclusions. In addition, the carrying position of the smartphone should be less important in order to allow more flexible data collection.

Future work should make all important information available to facilitate further work. This includes the data of the study participants, the source code, the methods and algorithms used. The authors emphasize that smartphones are not only useful for detecting everyday activities, but also have the potential to influence public health at the population level.

The study populations were mostly small, often less than 30 participants, which limits the generalizability of the results. They argue that future research should focus on improving data quality and including a broader participant base to increase the applicability and reproducibility of the results.

In summary, the review shows that smartphones are a promising tool for health research, especially for the detection and monitoring of physical activity. However, further efforts are needed to optimize the collected data, analysis methods and to apply the research results to larger and more diverse populations. [SJO21]

2.3.4 Accuracy of Smartphone Applications and Wearable Devices for Tracking Physical Activity Data

The research letter "Accuracy of Smartphone Applications and Wearable Devices for Tracking Physical Activity Data" by Case et al. examines the accuracy of smartphone apps and wearable devices for recording step counts in comparison to direct observation.

Healthy adults were recruited to walk 500 and 1500 steps on a treadmill. A total of 14 participants took part in 56 walking trials. The devices included the Digi-Walker SW-200, Fitbit Zip, Fitbit One, Fitbit Flex, Jawbone UP24, Nike Fuelband and the apps Fitbit, Health Mate and Moves on an iPhone 5s and a Galaxy S4.

The results showed that pedometers and accelerometers recorded step counts very accurately, with a deviation of -0.3% to 1.0%. Wearable devices had larger deviations, especially one device that counted up to 22.7% fewer steps. Smartphone apps showed a deviation of -6.7% to 6.2%.

The results were broadly consistent for both step lengths (500 and 1500 steps). The results were broadly consistent at both step lengths (500 and 1500 steps) and the study indicated that many smartphone apps and wearable devices are accurate for recording step counts. It was performed with young, healthy volunteers in a controlled environment, which may limit the generalizability of the results.

Further studies in different settings and with different devices are needed to confirm the results. The authors argue that the use of these devices could lead to an increase in physical activity, which would result in health benefits.

This has not yet been established because pedometers are not widely used. These findings could increase confidence in the use of smartphone apps and wearable devices to monitor health behaviors and thus have important implications for strategies to improve public health. [CBVP15]

2.4 Previous Research on Step Counts

2.4.1 How Many Steps/day are Enough? For Adults

The article "How Many Steps/day are Enough? For Adults" by Tudor-Locke et al. deals with the question of how many steps per day are enough for adults to benefit their health. The authors analyze various recommendations and studies that deal with the measurement and interpretation of the daily number of steps.

Around the world, recommendations for physical activity have traditionally been based on the frequency, duration and intensity of exercise. These recommendations can be more easily reflected in daily step counts, which are made possible by pedometers and accelerometers. The studies were analyzed with a focus on healthy adults between the ages of 20 and 65 without chronic diseases or disabilities.

It was found that the average daily step count varies in different countries. In the USA, for example, it is around 6,500 steps per day, while the average in Japan is 7,200 steps per day. New measurement methods make it possible to objectively record physical activity and share this information publicly.

Studies have shown that healthy adults walk between 4,000 and 18,000 steps a day on average. A good target is 10,000 steps per day. Intervention studies, which investigate the effects of targeted measures, demonstrate that such interventions can lead to an increase of approximately 2,000 to 2,500 steps per day.

The authors defined moderate-intensity exercise as taking approximately 100 steps per minute, which equates to 3,000 steps in 30 minutes. It is important to note that these steps should be in addition to normal daily activity Current public health guidelines recommend a minimum of 150 minutes per week of moderate-intensity exercise to maintain good health.

There is also a recommendation for moderate to vigorous physical activity (MVPA), resulting in a daily step count of approximately 7,100 to 11,000 steps. Activities classified as MVPA include brisk walking, jogging, cycling, swimming, and similar exercises that demand more exertion compared to light activities such as leisurely walking or light household chores.

The authors provide a guideline for the minimum required MVPA of 7,000 to 8,000 steps per day. These advice should help all participants in future research and the general public to create better health messages and promote behavior change.

They express that recommendations based on step counts should align with current health guidelines. However, they emphasize that recommendations based on step counts should not replace existing variables such as duration and intensity, but complement them.

In addition, they point out that any extra physical activity is better than none, especially for inactive adults. [TLCB+11]

2.4.2 Daily Step Count and All-Cause Mortality: A Dose–Response Meta-analysis of Prospective Cohort Studies

The meta-analysis "Daily Step Count and All-Cause Mortality: A Dose-Response Meta-analysis of Prospective Cohort Studies" by Jayedi et al. examines the relationship between the number of daily steps and all-cause mortality in the general adult

population. A total of seven prospective cohort studies were analyzed and there were a total of 16,906 participants.

The mortality rate is a key indicator in epidemiological studies, representing the frequency of deaths in a given population during a specific period. The results of the analysis reveal that each 1000-step increase in daily step count is associated with a 12% lower mortality rate in adults and a 13% lower rate in adults over 70 years of age.

In the studies analyzed, a higher number of steps per day was found to be associated with a lower risk of all-cause mortality. This statistical correlation remains even after changes in some external factors. These include age, geographical location, the duration of the study and the devices used to count steps. The correlation could not be changed even when adjusting for confounding factors such as body mass index, alcohol consumption, smoking status, step intensity and pre-existing medical conditions.

A non-linear dose-response analysis found that 10,000 and 16,000 steps per day were associated with a 56% and 66% lower mortality risk compared to 2,700 steps per day, respectively, with higher step counts showing a significantly greater reduction in risk.

The findings were rated as strong based on the GRADE methodology, indicating that a higher number of steps per day is likely to reduce mortality rates, with future research expected to confirm this conclusion with little alteration. Previous systematic reviews (Hallal et al., 2012 and Lee et al., 2012) have also reported an inverse relationship between the number of steps and mortality.

The authors emphasize that counting steps is a simple and effective way to promote public health. In general, public health deals with the overall population and general health deals with the health of the individual. The recommendation of 10,000 steps per day is supported by the data presented in this meta-analysis. However, further research is needed to determine this threshold more precisely and to better understand the underlying mechanisms.

The studies employed both accelerometers and pedometers, with accelerometers demonstrating a clearer and more distinct association with lower mortality rates. Furthermore, accelerometers not only capture the number of steps but also measure the intensity and duration of activities, allowing for a more precise assessment of an individual's overall physical activity. Additionally, they can measure body movement in three dimensions, enabling a more comprehensive evaluation of activity levels.

Studies that accounted for step intensity exhibited a weaker correlation with mortality compared to studies where intensity was not monitored. This suggests that the total number of steps may exert a greater influence on mortality outcomes than the intensity of individual steps. This analysis demonstrates that increasing daily step count significantly reduces the risk of mortality.

This underscores the importance of physical activity, particularly walking, as an accessible form of exercise for public health. Promoting increased daily step counts through a simple public health approach could significantly contribute to reducing mortality rates. The authors' meta-analysis demonstrates a clear association between daily step counts and mortality rates. Higher step counts are consistently linked to lower mortality rates across different population groups, highlighting the importance of daily physical activity. [JGSB22]

2.4.3 Global physical activity levels: surveillance progress, pitfalls, and prospects

In their article "Global physical activity levels: surveillance progress, pitfalls, and prospects", Pedro C. Hallal and colleagues examine global physical activity patterns and their surveillance, highlighting the progress, challenges, and future prospects. Significant progress has been made in monitoring physical activity among adults globally over the past decade. Two-thirds of WHO member countries now have data on their population's physical activity, which is a significant advancement. The data provide a detailed overview of global activity patterns and inactivity risks, which is crucial for developing national health policies.

Significant gaps remain, particularly due to the absence of continuous national monitoring systems. These systems primarily exist in highly developed countries, while they are lacking in regions such as Africa, the poorest parts of Latin America, and Central Asia. One-third of adults and four-fifths of adolescents do not fulfill the recommended public health guidelines for physical activity. In most countries, there are significant differences in the frequency of physical inactivity, with women being less active than men and older adults being less active than younger adults. These clear patterns should help politicians to develop and implement effective programs for the prevention and treatment of non-communicable diseases.

Although occupational physical activity is declining in high-income countries, leisure-time activity has increased among adults. The WHO STEPwise approach, which is a method for systematically collecting health data to combat non-communicable diseases, provides a good basis for surveillance in low- and middle-income countries. Advances in technology and new measurement methods, particularly the use of accelerometry, offer promising opportunities for future physical activity monitoring if costs decrease and technology advances.

The International Physical Activity Questionnaire (IPAQ) and the Global Physical Activity Questionnaire (GPAQ) support national monitoring and allow a comparative assessment of global activity patterns. Worldwide, 31.1% of adults are physically inactive, with the highest proportions reaching approximately 43% in the Americas and the Eastern Mediterranean region, and the lowest proportions at 17% in Southeast Asia. The frequency of inactivity increases with age and is higher among women than men, while it increases in high-income countries.

Additionally, 80.3% of adolescents aged 13 to 15 do not meet the recommended level of physical activity, with boys being more active than girls. The technological advancements since the Industrial Revolution have enabled people to reduce physical labor. Engines and vehicles exert conspicuous effects on physical activity, whereas technologies such as televisions and computers manifest subtler and more intricate impacts.

The authors contend that utilizing existing data for monitoring physical activity in the 21st century poses a significant challenge, as many individuals are becoming less active compared to previous times. The traditional approach to public health, which relies on evidence and persuasion, has so far proved largely ineffective. The authors' findings indicate that global surveillance of physical activity is advancing due to governments and international organizations being prompted by the increasing burden of non-communicable diseases on healthcare systems. [HAB+12]

2.4.4 Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy

The article "Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy" by Lee et al. deals with the effects of physical inactivity on major non-communicable diseases worldwide. The authors I-Min Lee, Eric J Shiroma, Felipe Lobelo, Pekka Puska, Steven N Blair and Peter T Katzmarzyk analyze the burden of disease and life expectancy associated with physical inactivity. The study was prompted by the strong evidence that physical inactivity increases the risk of many health problems, including coronary heart disease, type 2 diabetes and breast as well as bowel cancer. In addition, physical inactivity shortens life expectancy, which is caused by a high inactivity rate in the world population.

The researchers aimed to quantify the impact of physical inactivity on significant non-communicable diseases. Their intention was to find out how many cases of disease could be avoided if inactive people became active and what impact this would have on life expectancy. The authors conducted a comprehensive analysis utilizing data from various epidemiological studies and health databases. They calculated the proportion of the population that is inactive and modeled how increased physical activity would impact disease rates and life expectancy.

The analysis revealed that a significant proportion of the global burden of disease is attributable to physical inactivity. Researchers estimated that approximately 6-10% of cases of coronary heart disease, type 2 diabetes, as well as breast and colon cancer are due to lack of physical activity. Moreover, they showed that life expectancy worldwide could increase by 0.68 years if physical inactivity were eliminated.

The results underscore the crucial role of engaging in physical activities to prevent major non-communicable diseases and to enhance life expectancy. Emphasizing the necessity of public health initiatives aimed at promoting physical activity within the population, the authors underscore that these measures could bring substantial health benefits and are crucial for improving population health worldwide. [LSL+12]

3. Research Questions and Hypotheses

3.1 Research Questions

The aim of this study is to investigate the effects of a self-developed mobile 3D-endless-runner exergame on users' daily step count. Both quantitative and qualitative data will be considered. This leads to several central research questions to be answered during the course of the study.

The main research question is: Does the use of a mobile 3D-endless-runner exergame significantly increases users' daily step count compared to a period without using the exergame? This research question is a central part of the investigation and simultaneously explains the primary goal of the study.

Several subordinate research questions will help answer the main research question. Firstly, how does the daily step count of participants change without and with the use of the exergame? This question aims to identify and analyze measurable impacts on physical activity. Another important question is: How does the exergame affects participants' motivation for physical activity? It will be examined whether and to what extent motivation changes over the 8-week testing period. Finally, the study will explore: What positive and negative usage experiences do participants report during the test phase with the exergame? This question allows for collecting qualitative feedback from participants, providing a comprehensive picture of their experiences and the user-friendliness of the exergame.

Answering these research questions aims to gain comprehensive knowledge on how mobile exergames can promote physical activity and the factors that play a role in this process. The research results of this study aim to contribute to the further development of exergames to improve their application in the health sector.

3.2 Hypotheses

Based on the formulated research questions and the objectives of the study, the following hypotheses are established. The main hypothesis is that the use of a mobile

3D-endless-runner exergame leads to a significantly increased daily step count of users compared to a period without using the exergame.

The subordinate hypotheses include several aspects. First, it is assumed that there are significant differences in the daily step count between the 4-week test phase without the exergame and the 4-week test phase with the exergame. The second subordinate hypothesis is that using the exergame increases participants' motivation for physical activity. Lastly, it is assumed that participants report predominantly positive and few negative usage experiences with the exergame.

These hypotheses, which form the basis for empirical investigation, allow the research questions to be systematically answered. This enables a targeted analysis of the data. With these data, the impact of the mobile 3D-endless-runner exergame on daily step count and participants' usage experiences can be better evaluated.

4. Methodology

4.1 Participants

The study included a total of two participants aged 24 and 27 years. The gender distribution was balanced, with 50% male and 50% female participants. During the study, participants were asked to maintain a diary throughout the 4-week testing phase, which included questions about any known health limitations. All participants confirmed that they had no known physical restrictions that could affect their participation in the study.

Inclusion criteria for participation in the study were a minimum age of 18 years and access to a stable internet connection.

Participant recruitment began in February 2024. In addition to the author, another volunteer was recruited to participate in the investigation.

Before the study commenced, all participants signed an informed consent form, indicating their willingness to participate in the study and were informed that their data would be treated confidentially.

4.2 Procedure

This study adopts a mixed-methods approach, allowing for the collection and analysis of both quantitative and qualitative data. This strategy leverages the advantages of both approaches, providing a comprehensive understanding of the app's usage through user experiences and step data.

The mixed-methods approach employs quantitative analysis of step data and qualitative analysis of the case study, facilitated by diary entries. This enables the capture and analysis of objective behavioral data and subjective user experiences.

The study follows a descriptive research approach, aiming to analyze the effects of using a mobile endless runner exergame on users' daily step counts. By examining the quantitative step data collected before and during the use of the exergame, causal

relationships between exergame use and users' daily step counts are identified. Additionally, the qualitative diary entries are analyzed in order to determine experiences and influencing factors during the use of the exergame.

The development of the 3D Endless Runner was an iterative process. Initially, concepts were developed and paper prototypes created. During this early phase, ideas were gathered, discarded, and redesigned. Practical tests were conducted on treadmills and in outdoor environments to identify safety issues and refine framework conditions. This process constrained the functionality of the application. Consequently, programming proceeded in stages until the game could be completed.

After the completion of the app development, the participants were informed about the study. Each participant signed a consent form and was briefed on potential risks.

As part of the study, participants were asked to move about their daily routines as usual over a period of four weeks. Only the step data was of relevance.

After the phase for collecting baseline data, participants were informed about the case study and the associated diary. The objective of the case study was to determine whether using the exergame led to an increased number of steps in daily life. Participants were asked to record specific information in their diaries, preferably on a daily basis. Following this briefing, participants were asked to undergo another four-week testing phase, this time integrating the exergame into their daily routines. No mandatory usage was enforced.

The collected step data was retrieved via the "Nocodeapi"[NoC24] interface and compared with the participants' diary entries. The aim was to identify patterns and correlations to gain insights.

Subsequently, the data was interpreted and analyzed to provide an overview. The step data was presented in tables and line graphs, with each participant having their own table and graph. The tables compared the step information both before and after using the exergame. The line graph displayed the participant's data both without and with the use of the exergame. Data without the exergame was shown in a lighter color, while data with the exergame was emphasized with a stronger color. The lines in the graph could intersect or even overlap. The statistical analysis of the data included arithmetic mean and median values.

Figure 4.1: Participant on the treadmill with the app

4.3 Variables

4.3.1 Framework conditions

Certain conditions are crucial for the safe use of the game. The application area should ideally be confined to a secure space, such as a restricted area or a treadmill. The purpose of these measures is to ensure user safety and minimize potential risks arising from uncontrolled environments.

Furthermore, there are conditions that limit the user's devices. The exergame was programmed exclusively for Android smartphones. Additionally, it is limited to modern Android devices. Versions below Android 5.1 "Lollipop" (API Level 22) cannot play or install the game. For development, the Unity engine was used in Unity version 2021.3.24f1, and the code was written in the C# programming language. The exergame was developed solely for Android smartphones, thereby excluding other operating systems such as iOS or Windows from using the game. The choice of the Unity platform and modern Android devices was made for several considerations. Newer versions of Android offer better security features and more performance, thereby enabling improved functionality and user experience. Older devices may struggle to display the game smoothly and with sufficient graphics. The selection of the Unity engine offers several advantages over others. It is a widely used and powerful development environment. Additionally, it allows the use of prefabricated assets and tools available in its own store. This prefabricated assets were also used in the creation of the app. Utilizing these prefabricated assets can accelerate the development process and enhance the quality of the end product. C# is primarily used for programming in Unity because it is optimized and facilitates easy integration with Unity APIs. This not only makes the application more efficient but also facilitates future maintenance and expansion efforts.

The pace at which the user moves and interacts with the game should be closely examined. Beginners, who lack experience using a treadmill, should initially start at a slow walking speed and increase it gradually. This recommendation primarily serves to prevent falls, which could result from a loss of control at higher walking speeds on the treadmill, especially when an additional distracting component, such as smartphone use, is involved.

Increasing distractions associated with hazards necessitate the restriction of the game's use in traffic. The player's focus on the game could lead to insufficient attention to traffic, thereby exposing them to significant dangers. This is why the game should only be used in a safe and controlled environment to avoid potential risks.

The ergonomic design of the game is another crucial aspect. It is designed to be operated with one hand, regardless of whether the user is left- or right-handed. The reason for this design choice is to account for unforeseen situations, such as falls, which can be more easily managed when the smartphone is used with one hand. This allows the user to maintain better control and minimize potential damage.

In terms of accessibility, the game presents both opportunities and challenges. The target audience encompasses a wide range of users, regardless of age, including young and old individuals as well as children. However, people with disabilities may encounter insurmountable obstacles when using the game. Particularly for wheelchair

users, who primarily rely on wheelchairs for mobility, the challenges are significant and the use of the application is less suitable, as movement detection is based on steps. The movements require changes in values along the Z-axis, which are difficult for wheelchair users to achieve. Since the interval for detecting steps is five seconds, it allows users enough time to take another step. This large interval is intended to accommodate seniors or individuals with mobility impairments, as they may not be able to move quickly. This allows for continuous step detection without interrupting the gameplay. Unfortunately, there is no way for blind users to use the game, as it is primarily visually based and lacks alternative control and information delivery mechanisms.

A central aspect is the application's privacy policies, which are responsible for the security and confidentiality of user data. Only the step data from a single day collected by Google [Goo24] and the status of whether steps are currently being taken, detected by the smartphone sensors, are recorded. This provides a general overview of the user's daily movement, which is not interpreted by the application. All step data from Google, previously provided by Nocodeapi [NoC24], are used only until the start of the next day and are then overwritten. The step data collected by Google are subject to Google's privacy policies. Nocodeapi reads these data and thus enables application access to them. Consequently, these data are also subject to Nocodeapi's privacy policies. Only users who have previously registered with Nocodeapi and logged in with Google can use this game and thereby agree to Nocodeapi's privacy policies. The data are used exclusively to determine the playable time in passive mode and to provide a general overview of the collected steps to the user in the main menu. The data indicating whether the user is currently active are determined by the smartphone sensors when the user has the game open in active mode. This information is collected, processed, and then overwritten every 5 seconds. It serves no other purpose than to detect movement and thus enables gameplay in active mode.

In summary, the described conditions provide a solid foundation for the use of the game. The focus is on ensuring safe, straightforward, and largely barrier-free usability to effectively mitigate potential hazards.

4.3.2 Objective Setting

Specific goals are pursued when using the game. It aims not only to provide fun and entertainment, but also to promote the user's health and daily activity level. Generally, exergames are employed to change user behavior in the long term, thereby integrating increased physical activity into daily life. Some exergames strive to achieve a daily activity level through an increased step count, aiming for a target of 10,000 steps per day to improve overall health. Additionally, daily engagement with the application is intended to be ensured by the enjoyment derived from playing the game.

The exergames described in the literature consist of a movement component and a gameplay component. They can significantly increase motivation for physical activity and offer health benefits. A central model responsible for the success of these games is Csikszentmihalyi's "flow" concept. This concept describes an optimal state that a user can experience during gameplay, where motivation and concentration are at

their peak. This state can be achieved when the user is in a perfect balance between their abilities and the challenges presented by the game.

According to studies, exergames are not only a means of increasing physical activity, but can also help to improve mental health. The physical activity regularly promoted by exergames can help to reduce the symptoms of stress and depression, making it a valuable tool in the prevention and treatment of mental disorders.

Endless runner games, such as the "ActivRunner" presented in the literature, allow deeper immersion into the game through motion detection. This is otherwise not possible in any other traditional game and simultaneously offers the possibility of physical activity. The use of smartphone sensors such as accelerometers and gyroscopes enables a broader range of motion, facilitating natural game control that enhances enjoyment.

Another motive for such games is their appeal to a broad spectrum of people. A study from the literature has demonstrated that even individuals with low physical activity levels benefit from exergames. They were motivated to increase their physical activity because the games were enjoyable and involved physical exertion.

In summary, the combination of exergames and endless runner games serves not only for entertainment but also concurrently improves users' health and enhances physical activity. These objectives can be more effectively achieved through the application of the "flow" concept and the utilization of modern technologies such as motion detection.

4.3.3 Game Structure

The game is divided into an active and a passive mode to provide an optimal gaming experience. The gameplay component of both modes consists of a 3D endless runner game, which is enhanced by certain functionalities. This structural division makes it possible to integrate active movement and the use of collected steps taken throughout the day into the gaming experience.

At the beginning, the game starts in the main menu, where the user can choose to play in either active mode or passive mode. After selecting the desired mode, the player controls an in-game character in the 3D endless runner game, which can move between two different lanes. The control is done by simply swiping on the smartphone display, supporting an intuitive gaming experience. From here, the user can access the pause menu. When exiting the pause menu, the user can either return to the game in one of the modes or go back to the main menu.

In active mode, the 3D endless runner game can only be played when the user is physically moving. Any movement of the player is detected by the smartphone's accelerometers, with changes along the Z-axis being particularly significant. Many games also use GPS data to detect movements. However, the use of GPS data can be very unreliable indoors or across different floors. For this reason, the app's movement detection is designed to rely solely on accelerometers. This design decision allows the game to be reliably used outdoors, indoors, in stairwells, or on a treadmill.

In passive mode, the 3D endless runner game can also be played, but with some differences from active mode. The gameplay time in passive mode is limited by the steps taken throughout the day. Therefore, a large part of this mode functions as an

idle game, where the user's steps are collected in the background by Google, even when the game is not open. As soon as the calculated playing time has expired, the game ends automatically. If the time runs out, the user can continue playing in active mode or collect more steps to reactivate passive mode.

With this clear distinction between the two game modes and precise movement detection, it enables a multi-faceted gaming experience. Additionally, flexible adjustments can be made to suit the individual gaming behavior of users. They can choose between physical movement in active mode or using collected steps in passive mode.

4.3.4 Step detection

The step detection is an essential component of the mobile application to accurately track users' physical activity and respond accordingly. Google and the internal sensors of the smartphone provide the necessary information for step detection. According to the literature, smartphone sensors are considered a reliable method to accurately measure users' step counts, deviating from the actual walked steps by a maximum of 6%. These data were determined in 2015 using devices available at that time. Modern smartphones are probably more accurate.

Activity data is continuously collected in the background by Google, accessible through the Google Fit API. The application utilizes Nocodeapi to extract a JSON file containing information from the last few days. These details include steps count, active minutes, calories expended, heart minutes, sleep segments, weight and activity summary. However, since step information is the primary focus of the app, additional queried indicators are not necessary. Additionally, only the step data for the current day is relevant as it is displayed on the user's main menu. Theoretically, it is possible to retrieve data over a 30-day period to establish trends.

The relevant time period starts at 00:00:00 AM and ends at 23:59:59 PM GMT+2. However, information on Google servers is stored in GMT format. Therefore, the predefined function "today" cannot be used. Consequently, querying Nocodeapi requires a specific time specification to output data in German GMT+2 format. Since Nocodeapi offers this functionality, it is possible to define a custom time period (customTimePeriod) in the query.

Each recorded step is converted into 0.18 seconds of gameplay time and then transferred to a time account. This means that 10,000 steps result in 30 minutes of gameplay time. Information about the time played and the time corresponding to the total number of steps is stored in containers. Every second played in passive mode decreases the time collected through steps. However, this gameplay time is not affected when the user plays in active mode. If the app is restarted, the information stored in the containers is retained. An important functionality is the resetting of the played time at the end of each day. This prevents the game from incorrectly assuming that the user has already played, which would otherwise result in the user needing to take more steps than the previous day to play again in passive mode.

The step data from Google primarily serves to provide an overview of users' daily activity and to enable playing in passive mode. The second component of step detection is real-time step detection, which is facilitated by the smartphone's accelerometers and is required for active mode. The sensors are used because the data

from Google, due to the large query interval, cannot provide real-time information for step detection.

When detecting steps in real-time through the sensors, the change in values along the Z-axis is particularly relevant. This reflects the vertical movement when the user holds the smartphone upright. Steps are checked every 5 seconds. If a certain threshold is exceeded, a step is detected. The number of steps is not relevant, only whether a step was made or not. If a step is detected, the game is continued, otherwise the pause menu is called up.

Accurate step detection is based on the integration of data from Google and real-time detection through smartphone sensors. This enables seamless integration of passive and active game modes, creating an immersive gaming experience that accurately reflects the user's movement.

4.3.5 Control of In-Game Characters

Precise control of the in-game character is of central importance for the user's gaming experience. The control tests and improves the player's skills and reaction time. The in-game character has several animations that are played based on the user's control. These include animations for jumping, lowering, and an idle animation that runs continuously. At the beginning of the game, the in-game character is positioned in the left lane and can be controlled through swipe gestures on the display. The control via tilt gestures did not prove effective during the testing phase, as there were inherent fluctuations after each step. Additionally, it would adversely affect the overhead view of the smartphone.

If the user swipes left or right on the display, they can influence the in-game character's running lane. If the character is in the right lane, the user can swipe left to switch to the left lane, and vice versa. During these movements, the character rotates in the respective running direction, making it appear as if they are looking forward-left or forward-right. If the in-game character is in the left lane, they cannot move further left, preventing the character from leaving the play area.

Not only horizontal but also vertical swipe gestures are recognized. Swiping up makes the in-game character jump to avoid obstacles in different lanes. The jump movement goes upwards along the Z-axis and then brings the character back down. This jump mechanism can only be executed when the in-game character is on the ground to ensure that the user cannot exploit the game mechanics. The jump is enabled by a variable called "Jumpforce", which adjusts the power of the jump. The jump height is limited by the combination of jump force and gravity. After the jump, the in-game character falls back to the ground due to gravity. The speed of falling or rising is influenced by the vertical velocity (verticalVelocity). If the user swipes down, the in-game character is moved to the ground faster than the normal descent from a jump. This functionality helps to avoid obstacles that are close together more easily, allowing the in-game character to jump again more quickly.

Furthermore, a combination of different movements is possible. Examples include jumping or lowering with a simultaneous lateral movement. The order of movements does not matter in this context. However, opposing movements such as jumping and lowering or a left and right movement cannot be executed simultaneously. Movements to the left and right cancel each other out and initiate the new movement. This

allows the in-game character to temporarily stay in the middle of the two tracks and avoid obstacles on both tracks without jumping. However, this game mechanism requires a high level of skill from the user, as the timing must be precise.

Opposing movements, jumping and lowering, can only be applied unilaterally at the same time. Jumping is only possible when the in-game character is below a certain height, preventing the user from bypassing game mechanics and remaining permanently in the air. The combination of jumping and lowering is possible, where the jump is interrupted, and the lowering is initiated immediately. Theoretically, the combination of lowering and jumping is also possible, but it makes no sense in gameplay, as the character would simply jump normally.

The control options, utilized through swiping on the smartphone, provide the player with an immersive and challenging experience. This requires both skill and strategic thinking, resulting in an engaging gameplay experience.

4.3.6 Level Generation and Construction

The fundamental framework of the 3D Endless Runner consists of level generation and structuring, which involves a complex arrangement of modules. This allows for a seamless and dynamic gaming experience. Each level is composed of various components that are connected in stages, resulting in the typical Endless Runner format. The modules consist of different elements. The entire level can be constructed from 14 different modules, along with an initial module that contains no obstacles or items.

Each module is equipped with a ground surface structured by lanes, cliffs, mushrooms, and tree trunks. These modules are visually delimited on both sides by cliffs composed of various components, forming a continuous large unit. However, these cliffs do not prevent the user from leaving the lanes. An in-game logic system checks the user's lane position to prevent them from exiting the game field. The character runs on lanes consisting of two brown dirt paths over a grass-covered ground. Tree trunk obstacles and mushroom items can appear at various intervals on the lanes. Mushrooms with white stems have the additional ability to appear not only on the lanes but also to float above them. This challenges the player to jump strategically to collect mushrooms while avoiding obstacles.

The level generation is automated through the LevelGenerator class, arranging modules to create a dynamically changing game environment. Initially, 6 modules are placed: 3 starting modules without obstacles or items to ease the user into the game, and another 3 selected from a pool of 13 different modules containing obstacles or items. Subsequently, random modules from the pool of 13 are continuously placed as the player progresses a specific distance, excluding the initial module. This approach enables the generation of a varied and unique track

The character appears to move forward, although it remains fixed in position, while the ground continuously moves towards the player. An illusion of movement is created by combining the character's running animation with the changing ground position. As the game progresses, the speed of the ground increases, forcing the player to avoid obstacles. Eventually, the player will be stopped by obstacles. The initial speed is -7 and can increase every 30 seconds, reaching a maximum speed of -25 after 9 minutes.

The precise level generation and continuous adjustment of game speed present an ongoing challenge for the user, which is engaging and meant to be mastered. This serves long-term motivation and aims to entertain the user. The level generation is intended to give the game its unique characteristic as an endless runner.

4.3.7 Obstacles and Items

The game environment is influenced by obstacles and items, which dynamically enhance the gameplay experience. They provide continuous challenges to the player, potentially intersecting their path and affecting their progress. One such obstacle is tree trunks, which, upon contact with the character, immediately end the game. In such cases, the user is returned directly to the main menu from either the active or passive game mode. This interruption of the game flow leads to the loss of the current game progress.

Not only obstacles but also items can intersect the player's path, offering positive effects on gameplay. One such item is mushrooms, which increase the user's score by one point upon collection. Upon contact with one of these items, they disappear from the screen. The achieved score is continuously compared with the high score. If the current score surpasses the high score, it is set as the new high score. High score information is stored in containers and remains accessible even after the app is restarted.

The integration of obstacles such as tree trunks and items like mushrooms adds depth to the game. It aims to encourage the player to continuously enhance their skills and achieve new levels of performance. The varying challenges and rewards motivate the player to engage repeatedly with the game, thereby improving their skills over time.

4.3.8 GUI elements

The design of GUI elements in a game is of central importance. They serve to provide the player with an intuitive and user-friendly environment, allowing them to fully concentrate on the game and immerse themselves deeper into the game world. This section will closely examine the structure and functionality of the GUI elements from the main menu and game scenes A.2.

The main menu consists of several elements that provide information and interaction options to the player. The most important component of the main menu is the display of the current day's step count in the center. This information is intended to motivate the player by visualizing their daily activity. The available passive mode playtime is displayed at the top left corner. This time is determined by the difference between the calculated time from accumulated steps and the already played time of the current day. At the end of each day, the available playtime is reset to enable a new gaming experience the following day. The current high score is shown at the top right corner, which remains preserved even after restarting the app. The high score serves as an incentive for the player to improve their performance. All this information is displayed using text fields, as they do not require direct interaction with the user but solely serve the purpose of conveying information.

In the lower middle section of the main menu, there are two buttons that allow the player to start the desired game mode. The first button, "Play while walking", puts

the player into active mode, where the player must stay in motion to play. The second button, “Play with collected time”, starts the passive mode, using the time accumulated from steps. In this mode, the player doesn’t need to be actively moving and can play while sitting or lying down. After selecting a game mode, the player is directed to the game scene. Buttons were chosen for this purpose because interaction with the user is required.

In the game scene, the user can view various elements that help them track their progress and navigate through the scene. At the center of the game scene is the playing field, which includes its boundaries, the in-game character, obstacles, and items. It’s important to note that all elements of the game scene are designed in a low-poly style. This includes the in-game character, obstacles, items, ground, path, and cliffs. This design choice was made purely for aesthetic reasons rather than functional ones, aiming to create a cohesive and visually appealing experience.

Both game modes feature a button at the upper left corner to pause the game. Pressing this button opens the pause popup. When opened, the current game is frozen, and the player has the option to return and continue later. If the player started the game in active mode, the game will also automatically pause if no steps are detected by the smartphone’s sensors. This ensures that the player remains active.

If the game is closed while it is paused through the pause popup, the player cannot resume from the frozen game state. Upon restarting the app, the user is returned to the main menu. If the accumulated time runs out, the game exits and returns to the main menu, with only the high score saved and the active game progress lost.

The current high score is also displayed in the game scene at the upper right corner. It increases by collecting mushrooms. This text field is solely for information purposes and does not allow direct interaction with the player.

In the pause popup, the user has two interaction options. The “Continue” button allows the game to resume and the popup to close. The “Main Menu” button returns the player to the main menu, losing the current game progress. Only the new high score information is saved.

The integration of GUI elements, including text fields, buttons, and popups, ensures a seamless user experience, enabling transitions between different game modes and scenes. Additionally, the game can be paused and progress can be permanently monitored. The clear and concise display of game information ensures optimal user-friendliness.

5. Data Collection

5.1 Step Count Measurement

In this section, the methodology for collecting and analyzing step data is described. The data are used to examine users' activity levels before and after using the exergame.

Step data are collected quantitatively, providing a comprehensive overview of users' activity levels. Objective behavioral data, specifically the daily step counts, are collected to assess participants' physical activity. These data are essential for analyzing physical activity.

Step measuring is divided into two phases. In the first phase, steps are monitored over a 4-week period without the use of the exergame. In the second phase, steps are monitored over another 4-week period, this time with the use of the exergame. This division allows for a direct comparison, enabling a better evaluation of the exergame's impact on users' activity levels.

The participants' step measuring is conducted using the Google Fit app. The app allows for background monitoring of step counts as long as it is installed and configured on users' devices. Step data can be accessed via the Google Fit API. It is important to note that live step detection by Android smartphone sensors does not provide detailed information on step counts and is thus not suitable for assessing activity.

For data analysis, the Nocodeapi platform is used. Nocodeapi can generate a JSON file containing a user's step data for the past 30 days. A JSON file is a common data format that can be easily read and processed by both humans and machines and is widely supported in most programming languages. This JSON file is generated and analyzed after both the initial 4-week phase without the exergame and the subsequent 4-week phase with the exergame. Step data are recorded in GMT format, while German time is GMT+2, which can lead to discrepancies. Additionally, each day is internally represented from 10:12 AM to 10:12 AM the following day.

These temporal discrepancies can be eliminated by assigning each day its own adjusted time period. However, this would require a separate JSON file for each day,

as the pre-configured 30-day query does not allow for adjustments. The discrepancies pose only a minor issue in the analysis since the focus is on the trend in step changes. Furthermore, no steps are lost as they carry over to the following day. Accurate step tracking in the correct time format is necessary only for precise step display within the exergame. This functionality is implemented to ensure that users always see the correct daily step count.

Through this approach, a detailed analysis of users' physical activity can be conducted, both with and without the use of the exergame. This allows for valuable insights into the effectiveness of the exergame.

5.2 Case Study

The empirical research strategy of the case study allows for a detailed and intensive analysis of individual cases. In this investigation, the qualitative diary entries of the participants are analyzed. These subjective experience reports provide valuable insights and complement the objective step data.

Participants were asked to document their daily experiences with the exergame in diary entries. These entries capture individual experiences and subjective impressions, which can then be analyzed. The use of diaries was implemented during the second 4-week test phase, which included the use of the exergame. Ideally, entries were to be recorded daily to ensure the data was continuous and up-to-date.

To maintain a realistic setting, participants were encouraged to carry on with their normal daily activities. There was no mandatory instruction to use the app, ensuring that the usage and recordings were authentic and unaltered. Additional usage on a treadmill or in a controlled environment was permissible and recommended so that the active mode of the exergame could also be utilized.

The information recorded in the diaries included: App used (Yes/No), Observations while using the app, Issues encountered, It was fun (Yes/No), and Additional thoughts on the concept. This structure facilitates a systematic collection of user experiences, making subsequent qualitative analysis easier.

As described in the previous chapter "Methodology" 4.2, the goal of the case study is to determine whether using the exergame leads to an increased number of daily steps. The combination of objective step data and subjective diary entries allows for a comprehensive assessment of the exergame's impact on participants' activity levels. These data help highlight potential issues and positive aspects of the exergame based on user experiences.

5.3 Data Analysis Plan

This section describes the statistical methods used to analyze the collected data. The aim of this analysis is to draw meaningful conclusions and describe the impact of the exergame on users' daily step counts. The data analysis plan is divided into data preprocessing, statistical analysis, trend analysis, qualitative analysis and the integration of quantitative and qualitative data.

5.3.1 Data Preprocessing

Data preprocessing facilitates easier analysis of the data. The step data from both phases (without and with exergame use) are extracted from JSON files and numerically represented and sorted in respective tables. Additionally, the diary entries are digitized. These quantitative and qualitative data enable improved analysis.

5.3.2 Statistical Analysis

Descriptive Statistics

The statistical analysis begins with the calculation of basic descriptive statistics (mean, median, standard deviation) to provide an overview of the step data in both phases. This helps identify general trends and variations in the data.

Comparative Analysis

Next, the comparative analysis is performed using paired t-tests. This allows for a comparison of the average daily step counts before and after using the exergame. This analysis determines whether there is a statistically significant increase in step counts due to the exergame.

Trend Analysis

Following this, trend analysis is conducted based on time series analysis. This identifies trends and patterns in daily step counts over the 8-week period. Moving averages and trend lines are also used to visualize changes over time.

Qualitative Analysis

Finally, the qualitative analysis is performed by thematically analyzing the diary entries. This identifies common themes and insights regarding the use of the exergame. These qualitative data help better interpret the quantitative results.

Integration of Quantitative and Qualitative Data

The final component of the data analysis plan is the integration of quantitative and qualitative data. Through mixed-methods analysis, the quantitative step data and qualitative diary entries are combined to provide a comprehensive overview of the exergame's impact. Cross-referencing allows for an understanding of statistical results with user feedback, determining whether the exergame affects physical activity.

By employing this data analysis plan, the effectiveness of the exergame in promoting users' physical activity can be evaluated. These insights facilitate refinement of the exergame and improvement of its impact on users' health and fitness.

6. Results

This section presents the results of the data analysis obtained through the application of the data analysis plan. The step data are examined in two phases (without and with the use of the exergame). Additionally, the qualitative diary entries of the participants were analyzed, providing a comprehensive picture of their experiences with the exergame.

6.1 Data Preprocessing

The step data from both phases were extracted from JSON files and represented in tabular form. The diary entries were digitized and edited only for clarity to ensure all entries are readable. The step data (A.3) and the participants' diary entries (A.5) can be found in the appendix.

6.2 Statistical Analysis

6.2.1 Descriptive Statistics

Using these data, key statistical metrics such as mean, median, and standard deviation were calculated to enable a detailed analysis of the step counts.

Statistic	**Participant 1 without app**	**Participant 1 with app**
Mean	8051	9103
Median	7541.5	8895
Standard Deviation	2041.83	2194.55
Statistic	**Participant 2 without app**	**Participant 2 with app**
Mean	8613	9320
Median	8358.5	9303.5
Standard Deviation	2266.46	2014.12

Table 6.1: Descriptive Statistics of Steps

The descriptive statistics show that participants on average took more steps when using the exergame. The standard deviation indicates some variability in daily step counts.

6.2.2 Comparative Analysis

For the comparative analysis, paired t-tests were used. This statistical method identifies significant differences between the means of two related groups. In this case, the same participants without and with the use of the exergame.

Participant 1: The calculated t-value is t = 1.6844, which is smaller than the critical t-value of 2.052 at 27 degrees of freedom. The p-value is approximately 0.103, which is greater than 0.05 (alpha level). Therefore, the difference in daily step count is not statistically significant.

Participant 2: The calculated t-value is 1.1249, which is also smaller than the critical t-value of 2.052 at 27 degrees of freedom. The p-value is approximately 0.271, which is greater than 0.05. Here too, the difference in daily step count is not statistically significant.

These results indicate that the increase in step counts for both participants is not significant, even though both participants on average took more steps after using the exergame.

6.2.3 Trend Analysis

The trend analysis of step counts over the 8-week period shows fluctuations in daily step counts for both participants. The use of moving averages and trend lines enhances the detection of long-term trends and patterns in the data, filtering out weekly fluctuations such as differences between weekdays and weekends. The following describes the results of the moving averages for each participant, considering the phases of use.

The moving average of step data for the first participant during the phase without using the exergame shows a consistent trend with a slight declining tendency. These data can also be seen in a line chart in the appendix A.5. The smoothing of the data filtered out weekly fluctuations, indicating that the step count for Participant 1 remained constant but slightly declined. The minor decrease in trend could be attributed to natural fluctuations or other external factors, causing no significant change over the period.

In the second phase for Participant 1, where the exergame was used, the moving average is generally higher and shows a constant line. This line chart can also be found in the appendix A.6. Weekly fluctuations were also observed and smoothed out by the trend line. This constant line indicates that the use of the exergame did not cause significant changes in the long-term step count. The higher baseline suggests that the use of the exergame led to a general increase in step count despite stable trends.

For the first phase with the second participant without using the exergame, the moving average showed a consistent trend with a slight tendency to increase A.7. The smoothing of the data allowed for the elimination of weekly fluctuations, enabling a better examination of long-term development. The slight increase in activity could be attributed to natural improvement or other motivating factors.

The second phase, which included the use of the exergame, shows a significantly higher moving average. Although the values are notably higher, a declining tendency

can be observed A.8. Weekly fluctuations were also balanced. A higher average and a declining trend suggest that the step count significantly increased with the use of the exergame, but a certain decrease occurred over time. The initial increase in step counts due to the exergame is clear, but the declining trend indicates that the effects of the exergame diminished over time, possibly due to waning motivation or other external factors.

The moving averages and trend lines provide valuable insights into the long-term development of step counts. Although no statistically significant changes in step counts were found in the comparative analysis of the previous chapter, differences in trends for Participant 2 were identified. The data showed an increase in step count at the beginning of the exergame use, followed by a slight decrease. These observations offer additional contextual information for interpreting the data.

6.3 Qualitative Analysis

The qualitative analysis of the diary entries examines the user experiences with the exergame in more detail. These entries can be found in the appendix A.5. They offer valuable insights into the strengths and weaknesses of the app. Technical implementation posed challenges for some users. Certain functionalities of the exergame required prerequisites such as an installed Google Fit app and a stable internet connection, which were perceived as hindrance. Not every device comes with the Google Fit app pre-installed. Additionally, setting up the apps and logging in was complicated.

Users also reported limited variety and boredom after prolonged use of the exergame. The lack of variety caused by the same level being repeated contributed to the monotony, diminishing the gaming experience for users. Ergonomic and safety aspects were also mentioned, with the active mode perceived as potentially dangerous in daily use. Using the treadmill initially posed difficulties but became more comfortable with practice.

Another negative aspect was the loss of motivation. Users questioned the practical benefit of the high score in the exergame. Additionally, the obstacles that could always be overcome by jumping were seen as neither challenging nor varied. A component that users found missing was a visual display showing the progress of step counts over time. This feature could contribute to increased motivation.

The visual design of the app received mixed reviews. Some users found the poly graphics of the characters aesthetically pleasing, while others missed a corresponding poly background in the game mode and found the menus visually unappealing. The social isolation was also perceived negatively due to the lack of a feature to compare the high score with friends.

Physical limitations posed another problem expressed by participants. The exergame was not optimized for iPhones, creating additional effort for some users. Additionally, participants mentioned that not every user has a treadmill or a closed-off area at home, limiting the use of the active mode. Another frustrating component was the time limit during the game. The game abruptly ends when the user has no remaining playtime, which was particularly annoying when users were close to a new high score and could not continue the game.

Conceptual issues were another important aspect. Some obstacles in the game were hardly avoidable if logs were placed too close together. The high score achieved in the game could not be used to make progress. An introduction to the game explaining the mechanics was missing, and tailoring goals and playtimes to individual needs would have been beneficial.

Despite some criticisms, the basic concept of the game was positively received. The feature where the in-game character only moves when the user moves in real life was particularly motivating. Although there was some criticism of the high score, it was also seen as an incentive to continue playing. The accessibility and convenience of the passive game mode were praised, as it could be played anytime on the smartphone during breaks. Interaction with the app worked well, the controls were simple and reliable after some practice, and there were no crashes reported.

In summary, the qualitative analysis of the diary entries revealed several pros and cons of the exergame. Positive aspects included motivation to exercise and integration into daily life. Areas needing improvement were also identified, providing valuable suggestions for further development of the exergame and opportunities to enhance future user experiences.

6.4 Integration of Quantitative and Qualitative Data

The integration of quantitative step data and qualitative diary entries allows for an assessment of the exergame's impact on users' physical activity. Combining both data sources offers deeper insights into the exergame's effects.

The quantitative data indicate a general increase in daily steps during the use of the exergame. Statistical tests did not find a significant increase in daily steps, though trend analysis graphically shows an increased step count at the beginning of exergame use. This increase is supported by qualitative data documenting initial high motivation and positive user experiences.

The diary entries revealed that users found the exergame motivating and accessible, particularly the passive mode. These positive feedbacks correlate with the initially observed increase in step counts. However, qualitative data also showed a decrease in motivation over time, which is reflected in the quantitative data showing a declining trend in step counts.

A recurring theme in the diary entries was the monotony of the game and the desire for more variety. Following an initial increase in step counts, there was a plateau. The qualitative data help to explain this. Users reported that there was a lack of visual feedback showing their activity progress over time. Implementing such a feature could help maintain motivation for longer periods.

The setup of the exergame also revealed technical and conceptual issues. The primary concerns among participants were the requirement of the Google Fit app and the setup of the app via Nocodeapi. These feedbacks provide valuable insights for future developments of the exergame as they present concrete points for improvement.

In conclusion, the integration of quantitative and qualitative data demonstrates that the exergame has the potential to increase users' physical activity. Initial motivation

was high. However, for long-term changes that incorporate user experiences, adjustments and expansions of the game are necessary. These adjustments should ideally be based on feedback from the diary entries.

A holistic view of the exergame's impact on users can be obtained. The results show that a combination of motivating elements, technical improvements, and user-friendly design can help sustainably promote physical activity.

6.5 Hypothesis Testing

This section presents and compares the results of statistical tests conducted to evaluate the formulated hypotheses. The primary hypothesis posited that the use of a mobile 3D endless-runner exergame would significantly increase daily step counts. Additionally, three subordinate hypotheses were tested.

The subordinate hypotheses were as follows: 1) There are significant differences in daily step counts between phases with and without the use of the exergame; 2) The use of the exergame increases motivation for physical activity; 3) Participants report predominantly positive and few negative usage experiences with the exergame.

The first subordinate hypothesis was tested using the t-test, which revealed that the daily step counts of participants did not significantly increase. For Participant 1, the t-value was 1.6844 ($p = 0.103$), and for Participant 2, it was 1.1249 ($p = 0.271$). Since both p-values are less than the critical t-value of 2.052, the differences in step counts are not statistically significant.

The second hypothesis, which concerned motivation, indicated that participants were initially more motivated at the beginning of using the exergame. However, this motivation decreased over time, attributed to the monotony and limited variety within the game.

Regarding the third hypothesis, which pertained to usage experiences, diary entries revealed predominantly positive feedback concerning the motivational effects and the integration of the game into daily life. Nonetheless, technical difficulties, lack of variety, and ergonomic concerns were also noted. These negative aspects, however, were less frequent than the positive experiences.

In summary, the exergame did not have a significant impact on daily step counts. Initial motivation was high but decreased over time. User experiences were predominantly positive, although technical and conceptual issues were encountered. To enhance long-term motivation and user experience, improvements to the exergame based on user feedback could be beneficial.

7. Discussion

7.1 Limitations

The investigation of the exergame revealed several limitations that could affect the transferability of the results. One of the central limitations is the relatively small number of participants. A more representative result could have been achieved with a larger sample size. Additionally, a broader variety of age groups and fitness levels would have been beneficial. Improved diversity could lead to more comprehensive and generalizable conclusions. Particularly relevant would be the consideration of participants' gaming experience, as younger individuals often have more experience with video games, which could facilitate their engagement with the exergame.

Another important aspect is the short duration of the study. A longer testing period would have been necessary to better capture the long-term effects and sustainability of the exergame on participants' step counts. Furthermore, the intensity of physical activity and the duration of movements were not recorded. Analyzing these parameters could provide more precise insights into participants' fitness before, during, and after using the exergame.

A technical issue pertains to the collection of step data. These data are provided by Nocodeapi, which in turn accesses Google Fit servers. The datasets collected by Nocodeapi deviate from real time. A day and thus a dataset for steps begin at 10:12 AM and end the following day at 10:12 AM GMT. This means that the time zone does not correspond to the German format, and the query times do not run from 0:00 AM to 0:00 AM GMT+2. Due to this deviation, no steps are lost as they are carried over to the next day, but a better-adjusted timeframe would be beneficial for drawing more accurate conclusions.

In its current state, the exergame is difficult to access for individuals with certain disabilities, such as wheelchair users or the visually impaired. This limitation significantly reduces the inclusivity of the game and restricts the user base. Another issue is that in the active mode, the game can be manipulated by repeatedly moving the smartphone up and down to simulate steps. Step detection is checked every five seconds, allowing players to fake movement without actually being active. However,

the five-second interval offers some flexibility for seniors and individuals with limited mobility. Specific measures to prevent game manipulation have not been implemented.

Furthermore, the exergame was developed exclusively for the Android operating system. Expanding development to other operating systems would have significantly increased the development effort and was not implemented due to time constraints. In passive mode, the game ends when the playtime accumulated through steps is exhausted. This decision was made to motivate users to engage in more physical activity. However, resetting the current game state can lead to frustration among users when no further time is available. An approach could be to implement a save or resume function to preserve the current game state.

In passive mode, the time accumulated from steps is used for playing. The calculation requires an internet connection as the step data is retrieved from Google Fit servers. If the user does not have an internet connection and opens the app, problems can arise. A solution would be to record and process step data locally. Additionally, the exergame was originally designed for use in daily life. Tests showed that using it on public sidewalks can be dangerous as players focus on the game instead of their surroundings. Even implementing regular pauses every ten seconds did not fully solve the problem and worsened the gaming experience as the game progress was interrupted.

Finally, it should be noted that the functionality of the score in the exergame was not implemented. This decision was made due to time constraints. Originally, the score was intended to be used to enhance power-ups, which were to be integrated into a future version of the game. These power-ups were to include a magnet, a shield, and a multiplier, to expand gameplay and increase motivation.

These limitations highlight the challenges and potential areas for improvement of the exergame. They provide additional points of reference for future research and developments to enhance the functionality, accessibility, and user experience of the game.

8. Conclusion

8.1 Summary

This study investigated the impact of a self-developed mobile 3D-endless-runner exergame on users' daily step counts. The primary objective was to determine if using this exergame would lead to a significant increase in daily steps. Participants' step counts were monitored over two 4-week phases: a control phase without the exergame and an experimental phase with the exergame. While the increase in step counts was not statistically significant, the average daily steps were higher during the exergame phase. Participants reported both positive and negative experiences, with initial motivation for physical activity being notably higher at the beginning. The study suggests that exergames have the potential to promote physical activity through engaging and interactive means. This potential can be enhanced if motivation is sustained over a longer period.

8.2 Practical Implications

The results of this study highlight the potential of mobile exergames in promoting physical activity. Despite the lack of statistically significant results, the observed increase in step counts and the positive initial user feedback indicate that such games can effectively encourage a more active lifestyle. The integration of exergames into daily life, health promotion programs, and fitness routines can help make physical activity more appealing. This can be particularly motivating for groups who find traditional exercise monotonous. The study also underscored the importance of technical and design aspects of the exergame in improving the user experience. For instance, visual progress feedback contributed to maintaining motivation.

8.3 Future Research

Future research should focus on larger sample sizes and longer study durations to more effectively evaluate the impact of exergames on participants' physical activity. Investigating the long-term effects and sustainability of increased physical activity

through exergames is crucial. Furthermore, motivation during gameplay can be enhanced through various game designs, difficulty levels, and reward systems, potentially leading to longer-lasting behavioral changes in users. Integrating social features such as leaderboards could also enhance the game's appeal and effectiveness. The primary goal is to maximize users' health, which could be achieved by adapting to the diverse needs of different target groups.

A. Appendix

A.1 GitHub-Repository with App

https://github.com/r0bby97/3D-Endless-Runner

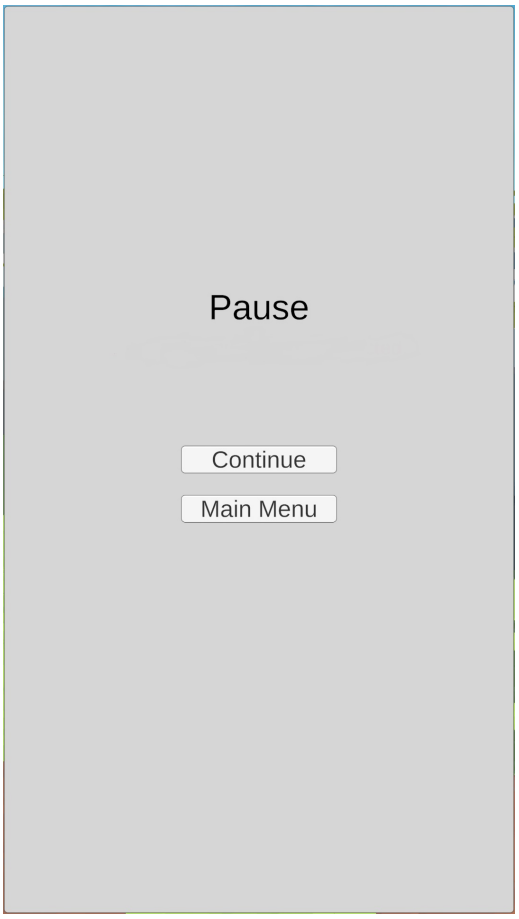
A.2 Ingame-Screenshots

Figure A.1: Ingame-Screenshot: Main Menu

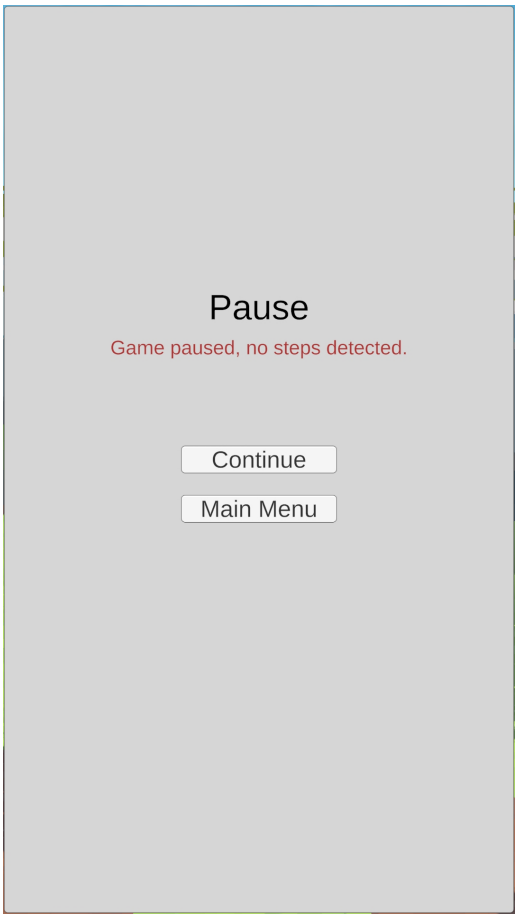
((a)) Active Mode

((b)) Passive Mode

Figure A.2: Ingame-Screenshot: Gameplay Modes



((a)) Normal Pause-Popup



((b)) Pause-Popup triggered by step detection

Figure A.3: Ingame-Screenshot: Pause Popups

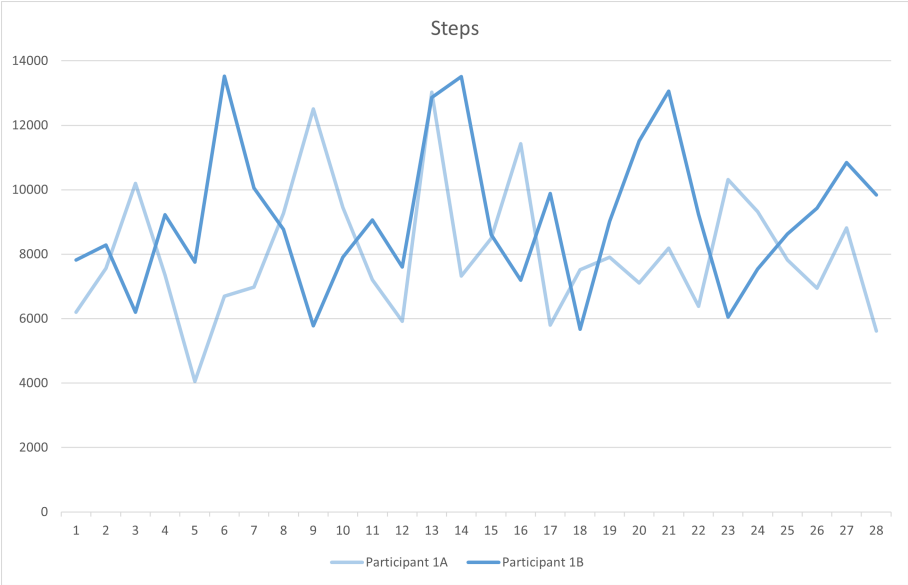
A.3 Participants' Step Data

Day	Participant without app	Participant with app
Day 1	6204	7821
Day 2	7560	8285
Day 3	10201	6196
Day 4	7359	9221
Day 5	4054	7752
Day 6	6701	13521
Day 7	6976	10052
Day 8	9267	8775
Day 9	12505	5782
Day 10	9456	7910
Day 11	7207	9066
Day 12	5923	7601
Day 13	13029	12866
Day 14	7321	13516
Day 15	8488	8608
Day 16	11430	7199
Day 17	5802	9881
Day 18	7523	5677
Day 19	7912	9015
Day 20	7107	11512
Day 21	8192	13062
Day 22	6380	9230
Day 23	10314	6052
Day 24	9327	7537
Day 25	7823	8626
Day 26	6946	9426
Day 27	8813	10843
Day 28	5620	9839

Table A.1: Steps of the first participant

Day	Participant without app	Participant with app
Day 1	7357	8921
Day 2	9802	9311
Day 3	12312	7737
Day 4	8634	10331
Day 5	6923	8555
Day 6	4370	12891
Day 7	8831	13558
Day 8	10247	9319
Day 9	13123	8311
Day 10	6032	6462
Day 11	7662	9635
Day 12	6820	11031
Day 13	7196	13511
Day 14	8392	9801
Day 15	6927	8202
Day 16	12523	10281
Day 17	10522	7633
Day 18	5427	9499
Day 19	6922	7982
Day 20	8325	12331
Day 21	7962	10910
Day 22	8644	6574
Day 23	13401	8522
Day 24	11003	9296
Day 25	6717	7157
Day 26	8830	10229
Day 27	9221	6022
Day 28	7044	6954

Table A.2: Steps of the second participant



((a)) Steps of the first participant

((b)) Steps of the second participant

Figure A.4: Comparison of steps of two participants

A.4 Moving Average

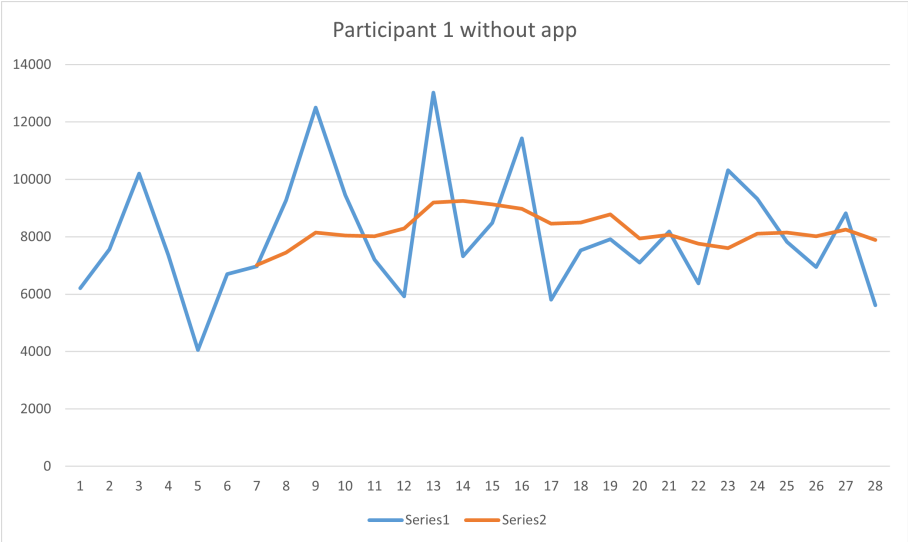


Figure A.5: Moving Average of Steps for Participant 1 without Exergame Usage

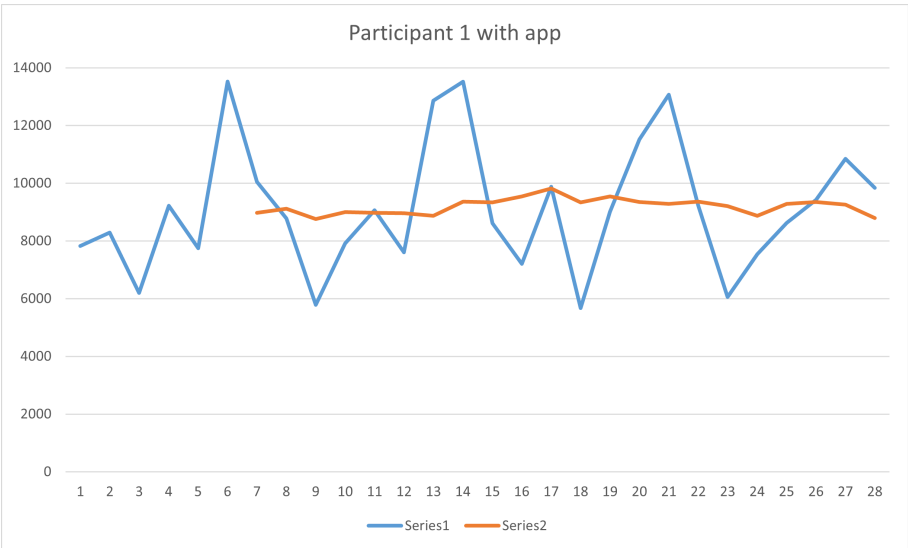


Figure A.6: Moving Average of Steps for Participant 1 with Exergame Usage

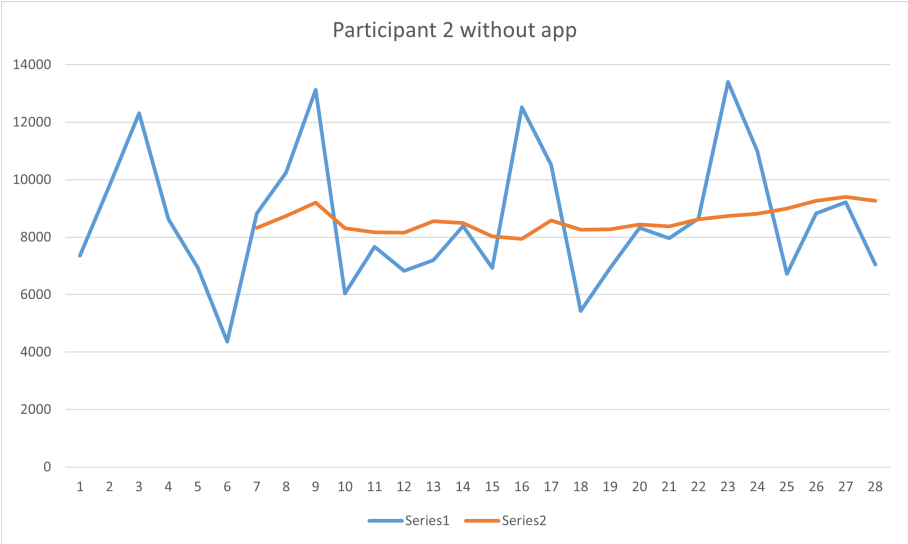


Figure A.7: Moving Average of Steps for Participant 2 without Exergame Usage

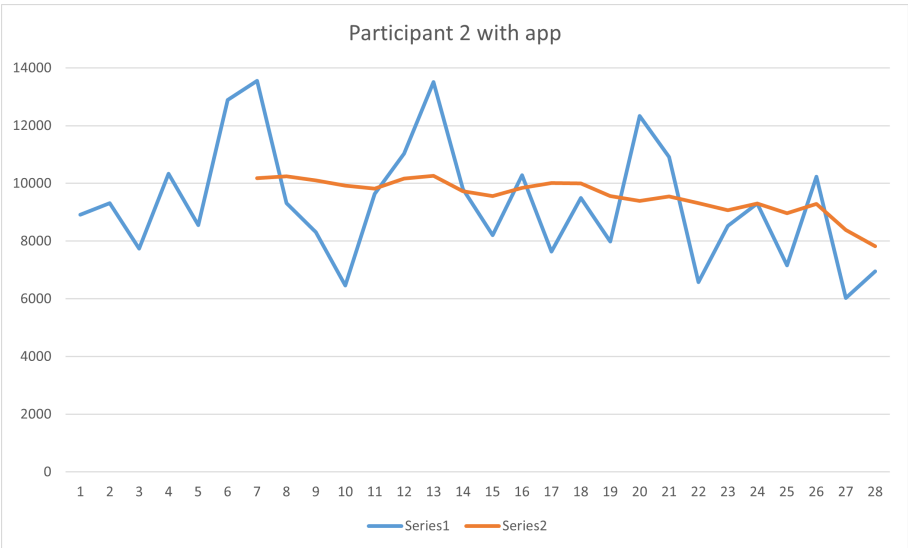


Figure A.8: Moving Average of Steps for Participant 2 with Exergame Usage

A.5 Participant Diary

Case Study - Nutzungsverhalten des Exergame-Anwenders (Participant 1)

☐ Gesundheitliche Einschränkungen, und wenn ja welche: _____

	App genutzt (Nein/Ja)	Das ist mir bei der Nutzung der App aufgefallen	Das hat nicht funktioniert	Es hat Spaß gemacht. (Nein/Ja)	Weitere Gedanken zum Konzept
01.04.2024	Ja	*1		Ja	*2
02.04.2024	Ja	Steuerung mit einer Hand klappt super		Ja	Optisch ansprechende Oberfläche wäre besser
03.04.2024	Ja	*3		Ja	
04.04.2024	Ja	Menü ist sehr eintönig gestaltet		Ja	
05.04.2024	Ja	Kurze Runden ermöglichen ein Spielen für zwischendurch		Ja	
06.04.2024	Ja	*4		Ja	*5
07.04.2024	Ja			Ja	*6
08.04.2024	Ja			Ja	
09.04.2024	Ja	Highscore motiviert mich ihn zu überbieten		Ja	
10.04.2024	Ja	*7		Ja	*8
11.04.2024	Nein				
12.04.2024	Ja	Die Schritte werden zuverlässig erkannt, *9		Ja	*10
13.04.2024	Ja	Hilft dabei, den Stress etwas zu vergessen		Ja	
14.04.2024	Ja			Ja	
15.04.2024	Nein				
16.04.2024	Ja			Ja	
17.04.2024	Ja	Highscorerun wird abgebrochen, wenn Zeit aufgebraucht		Ja	
18.04.2024	Ja			Ja	
19.04.2024	Ja	*11		Ja	*12
20.04.2024	Ja			Ja	
21.04.2024	Ja			Ja	
22.04.2024	Ja			Ja	
23.04.2024	Ja			Ja	*13
24.04.2024	Nein				
25.04.2024	Ja			Ja	
26.04.2024	Ja	*14		Ja	
27.04.2024	Ja			Ja	*15
28.04.2024	Ja	keine Abstürze der App		Ja	

*1 Das Spielen macht Spaß, da die Steuerung durch Gehen etwas Neues ist
*2 Sehr praktisch, dass nur ein Smartphone benötigt wird
*3 Die Schritte des Tages können gut im Hauptmenü verfolgt werden
*4 Falls man spielen möchte, aber gerade keine Zeit durch Schritte hat, dann kann nicht gespielt werden (nicht flexibel)
*5 Benötigt Internet um die Schritte aus dem passiven Modus zu bekommen, daher offline nicht spielbar
*6 Leider schwer im Alltag zu verwenden, da nur auf Laufband oder in einem abgesperrten Bereich
*7 Ich bewege mich gerne, da das Spielen spaß macht und ich einen höheren Highscore möchte
*8 Mit dem Highscore oder den gesammelten Pilzen kann man nichts machen
*9 Spiel wird automatisch pausiert, wenn ich mich nicht bewege
*10 Es wäre toll, wenn man die Highscores der Freunde verfolgen kann
*11 Die erspielte Zeit kann gut dafür genutzt werden um kurze Pausen zu überbrücken
*12 Eventuell mehr Level oder andere Hindernisse um für mehr Abwechslung zu sorgen
*13 Das Anzeigen einer Tendenz, wie sich die Schritte über einen Zeitraum verhalten (mehr Motivation)
*14 Meine Fähigkeiten im Spiel verbessern sich, wodurch ich höhere Highscores erreichen kann
*15 Für Personen, die sich mehr oder weniger bewegen kann die errechnete Zeit zum Spielen zu wenig oder zu viel sein (personelle Unterschiede)

Figure A.9: Diary - Participant 1

Case Study - Nutzungsverhalten des Exergame-Anwenders (Participant 2)

☐ Gesundheitliche Einschränkungen, und wenn ja welche: _____

	App genutzt (Nein/Ja)	Das ist mir bei der Nutzung der App aufgefallen	Das hat nicht funktioniert	Es hat Spaß gemacht. (Nein/Ja)	Weitere Gedanken zum Konzept
01.04.2024	Ja	Das Design des Menüs ist optisch nicht ansprechend	Einrichtung war schwierig	Ja	*1
02.04.2024	Ja	*2		Ja	Eventuell eine generelle Erklärung, was collected Time ist
03.04.2024	Ja	*3		Ja	Ich würde gerne den Score anderer Wissen
04.04.2024	Nein				
05.04.2024	Ja	*4		Ja	
06.04.2024	Ja	*5		Ja	Wovor rennt der Spieler weg?
07.04.2024	Ja	Teilweise schwer auszuweichen, wenn zu schnell		Ja	
08.04.2024	Nein				
09.04.2024	Nein				
10.04.2024	Ja	Keine Abwechslung, immer dasselbe Level		Ja	
11.04.2024	Ja		*6	Ja	
12.04.2024	Ja			Ja	Das kantige Design der Objekte im Spiel gefällt mit gut
13.04.2024	Ja			Ja	Warum ist das Design des Hintergrundes anders?
14.04.2024	Nein				
15.04.2024	Ja	h motiviert noch Einmal etwas mehr zu laufen und dabei		Ja	
16.04.2024	Ja			Ja	Nur auf dem Laufband verwenden ist nicht immer machbar
17.04.2024	Nein				
18.04.2024	Ja			Ja	
19.04.2024	Ja	*7		Ja	Was mache ich mit dem Highscore?
20.04.2024	Ja	*8		Nein	
21.04.2024	Ja			Ja	
22.04.2024	Ja			Ja	
23.04.2024	Ja			Ja	
24.04.2024	Ja			Ja	
25.04.2024	Nein				
26.04.2024	Ja			Ja	
27.04.2024	Ja			Ja	
28.04.2024	Ja			Ja	

*1 Verwendung auf iPhone wäre toll, Google Fit muss leider installiert sein
*2 Fast vom Laufband gefallen, weil ich mich zu sehr auf das Gehen auf dem Laufband konzentriert habe
*3 Die Steuerung ist sehr einfach und kenne sie aus anderen Spielen, Die Steuerung durch Gehen macht Spaß, Die Verwendung des Smartphones mit einer Hand ging gut
*4 Andere Spiele können auch Hindernisse durch Rollen ausweichen
*5 Die Anzeige auf dem Hauptbildschirm hilft mir dabei meine Schritte im Blick zu behalten
*6 Konnte im Modus "Play with collected time" nicht spielen. War im Zug und hatte kein Internet
*7 Highscore ist ein gute Idee um erneut zum Spielen zu motivieren
*8 Das Spiel wurde unterbrochen in meinem Highscore-Spiel und ich konnte dort nicht weiter machen

Figure A.10: Diary - Participant 2

A.6 Informed Consent Form

Einwilligungserklärung zur Studie

„The effects of a mobile endless runner exergame on users daily step count“

von Robert Lange

Durch die mündlich und/oder über digitale Kommunikationswege mitgeteilte Versuchsbeschreibung wurde ich über die Zielsetzung der Studie, den Ablauf des Experiments im Allgemeinen und über die eingesetzten Aufzeichnungsgeräte und Sensoren im Speziellen ausreichend informiert. Bei Bedarf konnte ich eine detaillierte, schriftliche Versuchsbeschreibung erhalten. Ich habe unmittelbar vor dem Unterschreiben dieser Einwilligungserklärung und der Versuchsdurchführung eine weitere mündliche Erklärung des Versuchsablaufs und der Datenerhebung erhalten. Ich habe die entsprechenden Informationen gehört/gelesen und verstanden. Meine mir wichtig erscheinenden Fragen wurden mir durch den Versuchsleiter zufriedenstellend beantwortet. Ich erkläre mich hiermit bereit, an der freiwilligen Untersuchung im Rahmen der oben betitelten Bachelorarbeit teilzunehmen.

Mir ist bekannt, dass während des Experiments Daten von mir gesammelt und auf elektronischen Datenträgern gespeichert und verarbeitet werden. Bei den gesammelten Daten handelt es sich um: Schrittdaten (zur Messung der allgemeinen täglichen Aktivität), Tagebucheinträge, sowie vom Versuchsleiter erstellte textuelle Notizen. Die Notizen dienen der Dokumentation von Beobachtungen, welche für die Evaluation der Studienergebnisse relevant sein können; etwa individuelle Besonderheiten bei der Durchführung bestimmter Gesten oder eventuelle technische Fehler und Probleme. Sie beziehen sich ausschließlich auf die für den Versuch und seine Evaluation relevanten Informationen.

Mir ist bekannt, dass die Teilnahme freiwillig ist und ich die Teilnahme jederzeit auch ohne Angabe von Gründen und ohne negative Folgen abbrechen kann.

Ich bin darüber informiert worden, dass es sich bei der hier durchgeführten Datenerfassung bei der Analyse der Schrittdaten nicht um eine medizinische oder klinischdiagnostische Untersuchung handelt und dass dementsprechend auch keine Informationen über meinen Gesundheitszustand abgeleitet werden.

Einwilligungserklärung zur Datenerhebung und -verarbeitung

Ich erkläre mich einverstanden, dass die im Rahmen dieser Studie erhobenen Daten von mir aufgezeichnet und auf elektronischen Datenträgern gespeichert werden. Alle Rohdaten, d.h. die Schrittdaten sowie alle Einträge im Tagebuch und Notizen werden ausschließlich unter Nutzung einer zufällig gewählten Versuchspersonennummer, d.h. pseudonymisiert, verarbeitet. Die Zuordnung der Versuchspersonennummer (die „Schlüsseldatei“) zur Personenidentität sind nur dem Versuchsleiter bekannt, werden von diesem verschlossen und verschlüsselt aufbewahrt und nur zu Zwecken der späteren Verifikation der erfassten Daten verwendet. Die Schlüsseldatei und die Schrittdaten werden spätestens bis zur Abgabefrist der Bachelorarbeit am 07.07.2024 gelöscht.

Figure A.11: Informed Consent Form - Participant 1

Ich bin damit einverstanden, dass die erfassten Schrittdaten, das ausgefüllte Tagebuch und die Notizen des Versuchsleiters zur Weiterverarbeitung und Analyse in einem internen, zugriffsgeschützten GitLab-Server des Cognitive Systems Labs (https://gitlab.csl.uni-bremen.de/) gespeichert werden. Der direkte Zugriff auf die Rohdaten ist abgesehen vom Versuchsleiter selbst nur für den Betreuer der Abschlussarbeit und die Gutachter*innen vorgesehen. Ich bin damit einverstanden, dass die Verarbeitung, Analyse, und Abbildung der anonymisierten Daten Teil der bis zum 07.07.2024 einzureichenden Abschlussarbeit sind.
Ich verstehe, dass ab dem Zeitpunkt der Einreichung der Abschlussarbeit mein Recht auf Auskunft über die in der Arbeit enthaltenen anonymisierten Daten, mein Recht auf Nutzungseinschränkung und mein Recht auf Löschung der Daten ggf. nicht mehr vollständig durchgesetzt werden können, sofern auf die Abschlussarbeit von Wissenschaftlern oder Studierenden abseits der genannten Personen und des Prüfungsamts des Fachbereichs 3 der Universität Bremen zugegriffen wird.
Ausgenommen davon sind die Schrittdaten und die Schlüsseldatei, welche auf einem separaten, während der Versuchsdurchführung nicht mit dem Internet oder den Versuchsgeräten verbundenen Rechners des Versuchsleiters aufgezeichnet, unmittelbar nach der Durchführung verschlüsselt und verschlossen, und nur dem Versuchsleiter selbst bekannt sein werden.

Ich kann jederzeit ohne Angaben von Gründen die vollständige Löschung meiner personenbezogenen Daten verlangen. Meine Teilnahme an dieser Studie ist freiwillig. Ich kann jederzeit während des Experiments, auch ohne Angabe von Gründen, meine Teilnahme widerrufen. Dadurch entstehen mir keine Nachteile. Die bis dahin erhobenen Daten werden in diesem Fall direkt unwiderruflich gelöscht.

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Figure A.12: Informed Consent Form (Continued) - Participant 1

Einwilligungserklärung zur Studie

„The effects of a mobile endless runner exergame on users daily step count“

von Robert Lange

Durch die mündlich und/oder über digitale Kommunikationswege mitgeteilte Versuchsbeschreibung wurde ich über die Zielsetzung der Studie, den Ablauf des Experiments im Allgemeinen und über die eingesetzten Aufzeichnungsgeräte und Sensoren im Speziellen ausreichend informiert. Bei Bedarf konnte ich eine detaillierte, schriftliche Versuchsbeschreibung erhalten. Ich habe unmittelbar vor dem Unterschreiben dieser Einwilligungserklärung und der Versuchsdurchführung eine weitere mündliche Erklärung des Versuchsablaufs und der Datenerhebung erhalten. Ich habe die entsprechenden Informationen gehört/gelesen und verstanden. Meine mir wichtig erscheinenden Fragen wurden mir durch den Versuchsleiter zufriedenstellend beantwortet. Ich erkläre mich hiermit bereit, an der freiwilligen Untersuchung im Rahmen der oben betitelten Bachelorarbeit teilzunehmen.

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Einwilligungserklärung zur Datenerhebung und -verarbeitung

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Figure A.13: Informed Consent Form - Participant 2

Ich bin damit einverstanden, dass die erfassten Schrittdaten, das ausgefüllte Tagebuch und die Notizen des Versuchsleiters zur Weiterverarbeitung und Analyse in einem internen, zugriffsgeschützten GitLab-Server des Cognitive Systems Labs (https://gitlab.csl.uni-bremen.de/) gespeichert werden. Der direkte Zugriff auf die Rohdaten ist abgesehen vom Versuchsleiter selbst nur für den Betreuer der Abschlussarbeit und die Gutachter*innen vorgesehen. Ich bin damit einverstanden, dass die Verarbeitung, Analyse, und Abbildung der anonymisierten Daten Teil der bis zum 07.07.2024 einzureichenden Abschlussarbeit sind.
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Figure A.14: Informed Consent Form (Continued) - Participant 2

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